53. Which one of the following statements best reflects the crux of the above passage?

(a) Environmental sustainability is inimical to our objective of achieveing a high rate of GDP growth.
(b) Poverty eradication is not possible without a rapid economic growth and the consequent environmental degradation.
(c) Maintaining high environmental standards is now a prerequisite for achieveing a steady, sufficient and inclusive growth.
(d) With large populations, rampant poverty and limited resources of today's world, environmental degradation cannot be prevented and inequalities are inevitable.

Ans. (c) : Options (a) is incorrect because it wrongly suggests that environment sustainability conflicts with achieving high GDP growth. However the passage advocate for rethinking growth strategies to integrate environment sustainability, not to oppose economic growth.

- Options (b) is incorrect, the passage, on the contrary, suggests that sustainable development strategies will help avoid an economic dead end and ensure inclusive growth.
- Options (c) is correct, The passage discusses the importance of high environment standards as a prerequisite for achieving sustainable.

Inclusive growth. It emphasizes the needs to rethink growth strategies to ensure they are environmentally sustainable while addressing the needs of poor and vulnerable groups.

- Options (d) is incorrect because it argues that environment degradation and inequalities are unavoidable due to current global situation. However the passage suggests the opposite, advocating for proactive measure like green growth to mitigate these issues.

54. A Question is given followed by two Statements I and II. Consider the Question and the Statements
Question: What are the unique values of x and y, where x, y are distinct natural numbers?
Statement–I : x/y is odd.
Statement–II : xy = 12
Which one of the following is correct in respect of the above Question and the Statements?

(a) The Question can be answered by using one of the Statements alone, but cannot be answer using the other Statement alone.
(b) The Question can be answer by using either Statement alone
(c) The Question can be answered by using both the Statements together, but cannot be answered using either Statement alone
(d) The Question cannot be answered even by using both the Statements together

Ans. (c) : Statement 1 : x/y is odd.
where x & y are the natural no.

$$\frac{x}{y} = \text{odd} \Rightarrow \frac{3}{1}, \frac{6}{2}, \frac{9}{3} \ldots\ldots\ldots$$

only from statement ⊥ we connot decide the values of x & y.
Statement 2 : x y = 12
if x = 6, y = 2
then $6 \times 2 = 12$
Hence, the question can be answered from both statements, but cannot be answered using either statement alone.

55. A Question is given followed by two Statements I and II. Consider the Question and the Statements.
A certain amount was distributed among X, Y and Z.
Question:
Who receivd the least amount?
Statement–I : X received 4/5 of what Y and Z together received.
Statement–II : Y received 2/7 of what X and Z together received.
Which one of the following is correct in respect of the above Question and the Statements?

(a) The Question can be answered by using one of the Statements alone, but cannot be answered using the other Statement alone.
(b) The Question can be answered by using either Statement alone.
(c) The Question can be answered by using both the Statements together, but cannot be answered using either Statement alone.
(d) The Question cannot be answered even by using both the Statements together.

Ans. (c) : Statement 1 :

$$x = \frac{4}{5}(y+z)$$

$$5x = 4y + 4z \ldots\ldots\ldots\ldots \text{(i)}$$

Statement 2:

$$y = \frac{2}{7}(x+z)$$

$$7y = 2x + 2z \ldots\ldots\ldots\ldots \text{(ii)}$$

Form eq^{n} (ii)

$$2z = 7y - 2x$$

Put in eq^{n} (i) –

$$5x = 4y + 2(7y - 2x)$$
$$5x = 4y + 14y - 4x$$
$$9x = 18y \Rightarrow x = 2y$$
$$\frac{x}{y} = \frac{2}{1} \ldots\ldots\ldots \text{(iii)}$$

Put x = 2y in equation (i)

$$5 \times 2y = 4y + 4z$$
$$6y = 4z$$
$$\frac{y}{z} = \frac{2}{3} \ldots\ldots\ldots\ldots \text{(iv)}$$

form eq^{n} (iii) & (iv)

$$x : y : z = 4 : 2 : 3$$

Amount of y will be least among these.
Hence options C is correct.

56. A Question is given followed by two Statements I and II. Consider the Question and the Statements.
Question:
If the average marks in a class are 60, then what is the number of students in the class?
Statement–I: The highest marks in the class are 70 and the lowest marks are 50.
Statement–II: Exclusion of highest and lowest marks from the class does not change the average.
Which one of the following is correct in respect of the above Question and the Statements?

(a) The Question can be answered by using one of the Statements alone, but cannot be answered using the other Statement alone.
(b) The Question can be answered by using either Statement alone.

(c) The Question can be answered by using both the Statements together, but cannot be answered using either Statement alone.
(d) The Question cannot be answered even by using both the Statements together.

Ans. (d) : Average = $\frac{\text{Sum of all number}}{\text{Total number of students.}}$

To calculate the average we need the total numbers of students and their marks. from statement 1 we cannot calculate the number of students because here only highest & lowest marks are given.
Statement 2:
Let's total no. of students are N.
according to given condition in statement 2,

$$\Rightarrow \frac{60N - 70 - 50}{N - 2} = 60$$

$$\Rightarrow 60N - 120 = 60N - 120$$

We are not able to find out the value of N. even after combining statement 1.
So, the question cannot be answered even by using both the statement together.

57. A Question is given followed by two Statement I and II. Consider the Question and the Statements.
There are three distinct prime numbers whose sum is a prime number.
Question: What are those three numbers?
Statement–I : Their sum is less than 23.
Statement–II : One of the numbers is 5.
Which one of the following is correct in respect of the above Question and the Statement?

(a) The Question can be answered by using one of the Statements alone, but cannot be answered using the other Statement alone.
(b) The Question can be answered by using either Statement alone.
(c) The Question can be answered by using both the Statements together, but cannot be answered using either Statement alone. (d) The Question cannot be answered even by using both the Statements together.

Ans. (a) : Prime numbers are→ 2, 3, 5,7, 11, 17.....,
Given– There are three distinct prime numbers whose sum is a prime number.
According to statement 1:
Sum should be less than 23.
if we take 3, 5, 11 prime numbers.
3 + 5 + 11 = 19 which is less than 23.
19 is prime no.
if we take other prime numbers
2 + 3 + 5 = 10 → not prime no.
According to statement of 1, there is one solution if we take 3, 5, and 11 as prime numbers. This can give the answer.
Statement 2: one of the number is 5
from statement (ii) we cannot find the exact solution of question because here is more than one conditions which will satisfied the condition which is given in question, as we can see –
3 + 5 + 11 = 19
13 + 5 + 11 = 29 and so on .....
We cannot find exact answer.
So, The question can be answered by using of the statement alone, but cannot be answered using the other statement alone.

58. A Question is given followed by two Statements I and II. Consider the Question and the Statements.
Question: Is (x + y) an integer?
Statement–I: (2x + y) is an integer.
Statement–II: (x + 2y) is an integer.
Which one of the following is correct in respect of the above Question and the Statement?

(a) The Question can be answered by using one of the Statements alone, but cannot be answered using the other Statement alone.
(b) The Question can be answered by using either Statement alone.
(c) The Question can be answered by using both the Statements together, but cannot be answered using either Statement alone. (d) The Question cannot be answered even by using both the Statements together.

Ans. (d) : x + y ⇒ an integer.
Statement – I:
2x + y ⇒ integer.
Check – $\frac{2}{3} + \frac{1}{3} = 1, \frac{1}{3} + \frac{1}{3} = \frac{2}{3}$,
It gives ⇒ {x + y = not integer}
Statement II:

$$x + 2y \Rightarrow \frac{1}{3} + \frac{2}{3} = 1$$

if we take $x = \frac{1}{3}$, & $y = \frac{1}{3}$ then both the statement satisfied but other than this value it will not give the correct answers.
The question cannot be answered even by using both the statements together.

59. A Question is given followed by two Statements I and II. Consider the Question and the Statements.
A person buys three articles p, q and r for ₹ 50. The price of the article q is ₹ 16 which is the least.
Question: What is the price of the article p?
Statement–I: The cost of p is not more than that of r.
Statement–II: The cost of r is not more than
Which one of the following is correct in respect of the above Question and the Statement?

(a) The Question can be answered by using one of the Statements alone, but cannot be answered using the other Statement alone.
(b) The Question can be answered by using either Statement alone.
(c) The Question can be answered by using both the Statements together, but cannot be answered using either Statement alone.
(d) The Question cannot be answered even by using both the Statements together.

Ans. (b) : From given information,
p + q + r = 50
q = 16, this implies p + r = 50 – 16 = 34
p, r > q = 16 this implies both p and r are at least 17
Question – what is the price of article p?
statement 1: cost of p is not more than r, p ≤ r
p, r can only 17, 17 which is valid as per this case, So, statement is sufficient to answer the question.
Statement 2: cost of r is not more than p, r ≤ p
p, r can only be 17, 17 which is valid as per this case so, statement 2 is sufficient to answer the question.
⇒ The question can be answered using either statement alone, hence options B is the correct answer.